



Design **THINKING**

UNIT 3

Dr. Ramakrishna Murthy K
EEE Department

SUMMARY

EMPATHIZE PHASE: Identifying with the user

- Be aware of the bias that you have
- Conduct interviews: Design broad questions, identify the end user, and conduct the interview
- Take down the observations
- Immerse yourself in the problem to identify the root
- Conduct research to gather more information about the design question
- Build an empathy map

DEFINE PHASE: Identifying the need

- Synthesize needs and insights: Group the observation and identify the needs. Structure it with cause of user behavior to get insight.
- Build an actionable problem statement with user, user need and insight
- Modify it to 'how might we' question



IDEATE

- Use imagination and creativity
- Creativity is the ability to go beyond existing ideas and create new ones
- “Creative confidence is like a muscle – it can be strengthened and nurtured through effort and experience”
- “Design is a sport where you have to actually participate”



IDEATE PHASE

Use what you've learnt from the empathy work to generate multiple solutions for the problem statement devised.

Step 1: Generate MANY ideas

Step 2: Choose the most intriguing and optimal solution

Demystifying creativity

Myth : Ideas are generated by **lone geniuses**



Did you think Thomas Edison sat alone in a garage and thought of solutions?

Demystifying creativity

Fact 1: Good innovators also need a team

Fact 2 : Nothing is completely original



**Thomas Edison
worked with a team
of 40 people in a
large-scale
research lab**



Topics for a Good Read

- Ford's assembly line

IDEO – Shopping cart challenge

www.youtube.com/watch?v=M66ZU2PClcM





IDEO – Shopping cart challenge

FRAMEWORK OF PRINCIPLES

- One conversation at a time
- Stay focused
- Encourage wild ideas
- Defer judgment
- Build on the idea of others

Pre-Brainstorming: Mindset, Warm-ups

- **Practical Optimism:** Positive thinking + Practical beneficial action intended to bring Good Results
- Knowing that **getting stuck is ok !**
- **Improvise, build on others' ideas:**
Try not to say 'Yes, **but**'.. Try saying 'Yes, **and**'
- "A good improviser is awake, not entirely self-focused, moved by a desire to do something useful, give something back and who acts upon this impulse"

Divergent thinking

- **Quantity better than Quality:** Do not try to fine tune your idea. Work on getting as many ideas as possible
- **Divergent Thinking involves:**
 - Encouraging wild ideas
 - NOT judge quality of ideas
 - Building on ideas of others
 - Getting unstuck



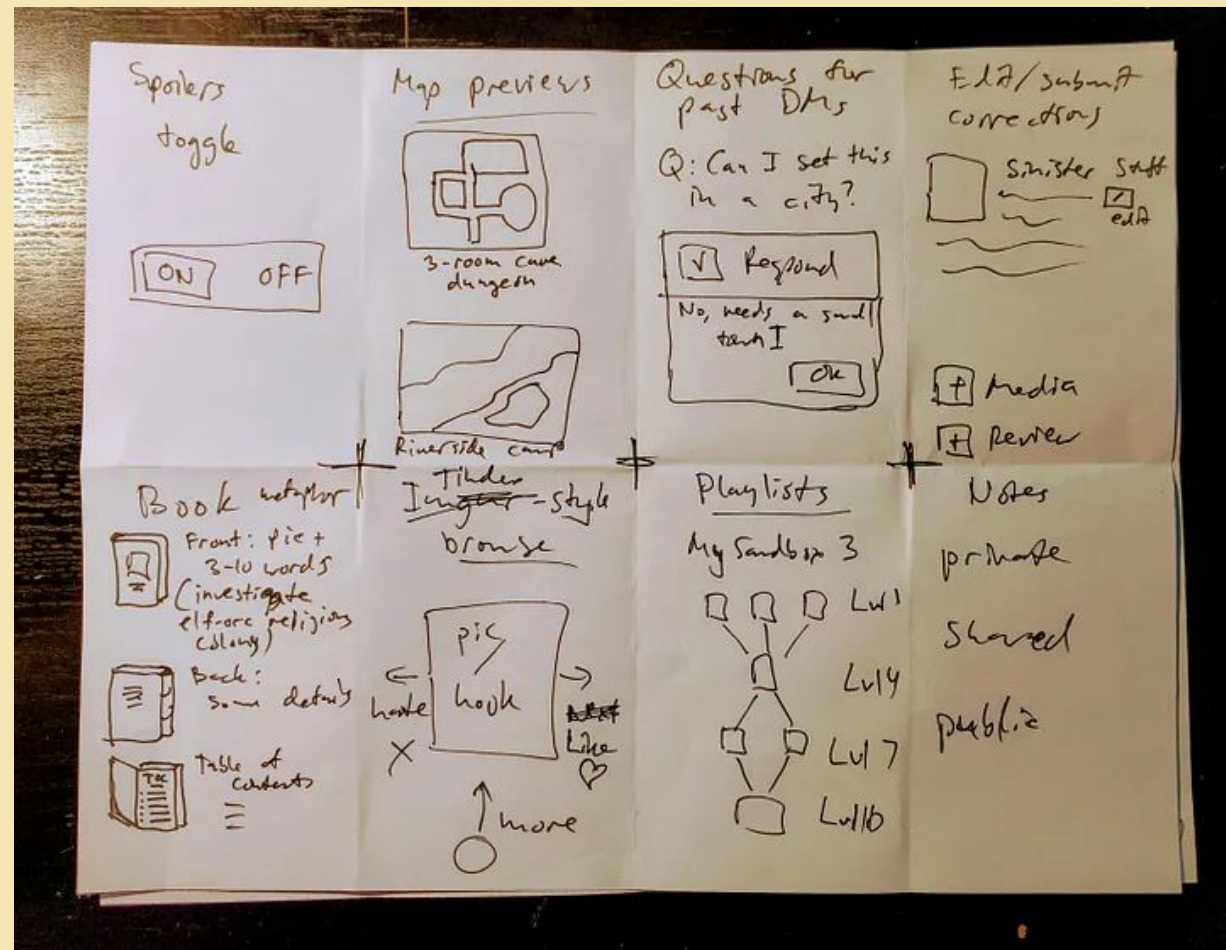
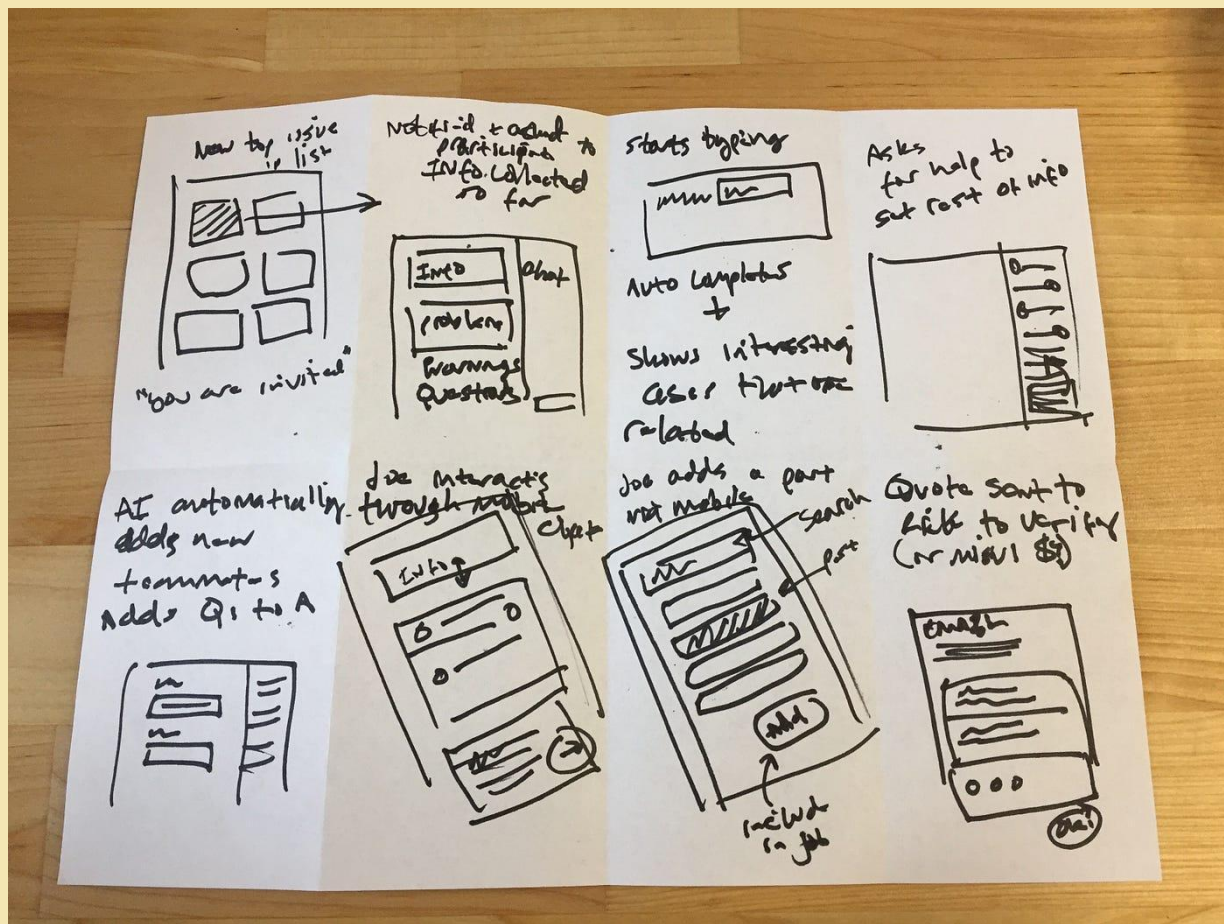
While ideating

- Try using sketches
- Try to suppress the urge to be perfect while drawing sketches

Note: Sketches are supposed to quickly and clearly communicate an idea drawn rapidly and roughly so that students can move on to the next idea

Building on ideas

- **Build by 'plus'sing on ideas of others:** See how you can add to others ideas (involves deferring judgment and using positive language)
- **Share your opinions with others:** Use sticky notes (or such) to share opinions
- **Getting unstuck:** Create subsidiary 'how-might-we' question or reframe problem statement.





Outcome of Divergent Thinking

- When you have enough ideas that appropriately addresses the problem statement, choose the best idea
- Best ideas are the ones that have the most innovation potential, or the capacity to lead to and produce innovative solutions to the design challenge

How to choose the Best ideas

Preferable Criteria:

- Most likely to make user happy.
 - Most likely to work.
 - Most likely to be interesting.
-
- If unable to narrow down as a team, VOTE for the best ideas by using “potential” criteria



After selecting Best Idea

- Make a blueprint

Note: Blueprint can be in the form of a technical drawings or strategic plan, a flow chart, a video blueprint, or so on.



Address the following in a blueprint

1. How does it benefit the end user and/or solve the problem statement?
2. Is the solution cool (new, different, special, unique, attractive, interesting) ?
3. Is the solution buildable?
4. How is the solution going to be tested so that it can elicit feedback and be improved?



Any Questions?